

ORIGINAL ARTICLE

Ventricular Arrhythmias in Marfan Syndrome: Prevalence and Clinical Correlates in a Multicenter Cohort

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BACKGROUND: Ventricular arrhythmias are increasingly recognized in Marfan syndrome, but associated clinical and structural features remain incompletely defined.

METHODS: We performed a retrospective multicenter cohort study of 783 adults with confirmed Marfan syndrome at 3 Mayo Clinic sites. Clinical, arrhythmic, echocardiographic, cardiac magnetic resonance, surgical, and implantable cardioverter defibrillator data were obtained by manual chart review. Associations with sustained ventricular tachycardia (VT) or ventricular fibrillation (VF) were evaluated using univariable logistic regression, a primary multivariable model in the full cohort, and a prespecified exploratory cardiac magnetic resonance sensitivity model.

RESULTS: Sustained VT/VF occurred in 74 patients (9.5%) over 12.7 ± 7.6 years. On univariable analysis, sustained VT/VF clustered with an adverse structural-electrical phenotype including atrial arrhythmias, ventricular remodeling, mitral valve disease, cardiac surgery history, and late gadolinium enhancement. In the primary multivariable model, nonsustained VT (odds ratio, 12.11 [95% CI, 6.43–22.79]; $P < 0.001$), lower left ventricular ejection fraction (odds ratio, 0.95 per 1% increase [95% CI, 0.93–0.97]; $P < 0.001$), history of at least 1 cardiac surgery (odds ratio, 3.59 [95% CI, 1.65–7.84]; $P = 0.001$), and moderate-severe mitral regurgitation (odds ratio, 2.00 [95% CI, 1.07–3.75]; $P = 0.030$) were independently associated with sustained VT/VF. In the cardiac magnetic resonance sensitivity model, nonsustained VT, lower left ventricular ejection fraction, and late gadolinium enhancement remained associated with sustained VT/VF.

CONCLUSIONS: In adults with Marfan syndrome, sustained VT/VF occurs in nearly 1 in 10 patients and is associated with a broader structural-electrical phenotype. Late gadolinium enhancement provided additional risk signal among patients undergoing cardiac magnetic resonance.

GRAPHIC ABSTRACT: A [graphShic abstract](#) is available for this article.

Key Words: atrial fibrillation ■ Marfan syndrome ■ mitral valve prolapse ■ population ■ tachycardia, ventricular

Marfan syndrome (MFS) is a heritable connective-tissue disorder caused by pathogenic variants in FBN1, most often recognized for its cardiovascular complications involving progressive aortic dilation and dissection.^{1,2} Although aortic events remain the principal cause of morbidity and mortality, growing evidence suggests that myocardial and electrophysiological abnormalities also contribute meaningfully to the disease

burden.^{3,4} Left ventricular (LV) dysfunction, mitral valve prolapse (MVP), and ventricular arrhythmias are increasingly described as part of the broader cardiac phenotype of MFS.

Multiple studies have reported a higher prevalence of ventricular tachycardia (VT) and ventricular fibrillation (VF) in patients with MFS compared with the general population.^{4–7} Ventricular arrhythmias occur in a minority but

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Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/CIRCEP.126.014967>.

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Circulation: Arrhythmia and Electrophysiology is available at www.ahajournals.org/journal/circep

WHAT IS KNOWN?

- Ventricular arrhythmias and myocardial involvement are increasingly recognized in Marfan syndrome, but their prevalence and associated clinical and structural features remain incompletely characterized.
- Prior studies have generally been limited by small cohorts, single-center designs, and incomplete integration of rhythm, echocardiographic, cardiac magnetic resonance, and device data.

WHAT THE STUDY ADDS

- In this multicenter cohort of 783 adults with Marfan syndrome, sustained ventricular tachycardia or ventricular fibrillation occurred in 9.5% of patients.
- Nonsustained ventricular tachycardia, lower left ventricular ejection fraction, history of cardiac surgery, and moderate-severe mitral regurgitation were independently associated with sustained ventricular arrhythmias.
- Among patients who underwent cardiac magnetic resonance imaging, late gadolinium enhancement provided an additional, hypothesis-generating risk signal.

Nonstandard Abbreviations and Acronyms

AF	atrial fibrillation
CMR	cardiac magnetic resonance
ICD	implantable cardioverter defibrillator
LGE	late gadolinium enhancement
LV	left ventricular
LVEF	left ventricular ejection fraction
MAD	mitral annular disjunction
MFS	Marfan syndrome
MR	mitral regurgitation
MVP	mitral valve prolapse
NSVT	nonsustained ventricular tachycardia
VF	ventricular fibrillation
VT	ventricular tachycardia

clinically important proportion of patients, with sustained VT or VF documented in up to 8% of cases. Recognition of these rhythm disturbances is clinically important, as they may contribute to morbidity and sudden death even in patients without advanced aortic disease. Proposed mechanisms include myocardial fibrosis, valvular abnormalities such as MVP or mitral annular disjunction (MAD), and altered ventricular repolarization dynamics.^{3,8,9} These findings suggest that structural and electrical remodeling may occur in MFS independently of aortic pathology.

Despite this emerging evidence, prior investigations have been limited by small sample sizes, heterogeneous diagnostic criteria, and incomplete imaging or

electrophysiological characterization. Most published studies have been single-center and lacked multivariate analyses or longitudinal outcome data. Collectively, these observations highlight the need for comprehensive evaluation of ventricular arrhythmias in MFS using contemporary imaging and clinical data. Unlike prior single-center series, the present study represents the largest systematically characterized cohort to evaluate both imaging-based correlates and device outcomes in MFS.

To address these gaps, we conducted a multicenter study of adults with MFS to define the prevalence, characteristics, and clinical correlates of sustained VT and VF. The study also evaluated the frequency, indications, and outcomes of implantable cardioverter defibrillator (ICD) therapy, including device-related complications. By integrating clinical, echocardiographic, and cardiac magnetic resonance (CMR) imaging data, the present study aims to clarify the arrhythmogenic profile of MFS and inform risk stratification in this population.

METHODS

Study Population

The individual participant data, analytic code, and study materials used in this study will not be made publicly available because of patient privacy concerns and the risk of reidentification in this retrospective cohort.

This retrospective cohort study included patients diagnosed with MFS between 2018 and 2024 at Mayo Clinic sites in Rochester, Arizona, and Florida. Patients were identified using the *International Classification of Diseases, Tenth Revision (ICD-10)* code, and all diagnoses were confirmed through manual chart review. A total of 1113 patients were identified through ICD-10 codes, of whom 783 remained after chart review confirmed the diagnosis.

A diagnosis of MFS was considered confirmed either through genetic testing demonstrating likely pathogenic or pathogenic mutations or through phenotypic criteria based on the Ghent criteria. Patients diagnosed phenotypically were assessed by a specialist, with diagnostic features including aortic dissection, ectopia lentis, or characteristic features in the presence of a family history of genetically confirmed MFS. Only patients in whom the diagnosis could be clearly verified by chart review and electronic health record documentation were included. Individuals younger than 18 years were excluded.

The study was approved as exempt by the institutional review board, and informed consent was waived due to the retrospective design. For longitudinal follow-up, the index date was defined as the first documented Mayo Clinic visit in the electronic health record. The last follow-up date was defined as the latest available Mayo Clinic clinical documentation, procedure date, or imaging date. Follow-up duration was calculated from the index date to the last available follow-up date.

Data Collection

Comprehensive chart review was performed to collect demographic characteristics, comorbidities, vascular history, surgical history, arrhythmia history, echocardiographic variables, CMR

imaging findings, and ICD data. All data were obtained by manual review of the electronic health record and were therefore dependent on available clinical documentation, imaging reports, rhythm tracings, and device interrogation records.

Detailed information related to VT and VF was abstracted, including event date, detection source, and underlying cause when identifiable. Nonsustained VT (NSVT) was defined as at least 3 consecutive ventricular beats lasting <30 seconds and terminating spontaneously. Sustained ventricular arrhythmia was defined as sustained VT and VF documented by chart review. Only arrhythmic events that could be clearly verified from the medical record were included. When rhythm strips, electrocardiograms, telemetry tracings, ambulatory monitoring reports, or device interrogation reports were accessible, morphology and heart rate characteristics were adjudicated when clearly confirmable.

Arrhythmias were also categorized based on their timing relative to cardiac surgery. Events that occurred intraoperatively or immediately postoperatively were not considered unless the arrhythmia was recurrent. For analytic purposes, cardiac surgery history was defined as surgery occurring before sustained VT/VF among patients with events and before last follow-up among patients without sustained VT/VF.

Ventricular arrhythmias were ascertained from available electrocardiograms, inpatient telemetry, ambulatory rhythm monitoring, cardiac arrest documentation, and ICD interrogation reports. Because this was a retrospective study, rhythm monitoring was not standardized across the cohort, and arrhythmia detection depended on clinical testing performed during routine care and on what was documented in the electronic health record.

Additional arrhythmia data included atrial fibrillation (AF), atrial flutter, supraventricular tachycardia (SVT), and conduction abnormalities such as left bundle branch block and right bundle branch block. AF subtype, when available, was categorized as paroxysmal or persistent, and antiarrhythmic or rate-control medication use was also abstracted when documented.

Echocardiographic and Structural Variables

Echocardiographic variables were abstracted from available clinical imaging and were not uniformly obtained before the first sustained VT/VF event. Therefore, these variables were analyzed as structural correlates of documented sustained VT/VF rather than as temporally established predictors. Specific parameters collected included left atrial volume index and left ventricular ejection fraction (LVEF). When available, 3-dimensional EF assessments were used; otherwise, 2-dimensional biplane methods were applied. Special attention was given to preintervention echocardiograms, those obtained before any valvular or aortic procedures.

Valvular abnormalities were systematically documented, including mitral, tricuspid, and aortic regurgitation, as well as bicuspid aortic valve. The severity of both regurgitation and stenosis was also recorded. Given the known association between MVP, connective tissue disorders, and arrhythmias, all patients were specifically assessed for the presence of MVP. Echocardiograms for all patients with MVP were reviewed to determine the presence of MAD. Patients without MVP were considered to have negative MAD. Cases with indeterminate echocardiographic findings were classified as missing data.

Additional structural variables abstracted from echocardiography included elevated left atrial volume index, abnormal LV diastolic dimension, abnormal LV systolic dimension, LV mass index, moderate-severe mitral regurgitation (MR), MVP, MAD, moderate-severe aortic regurgitation, bicuspid aortic valve, and moderate-severe tricuspid regurgitation.

CMR Imaging

CMR imaging data were also reviewed for all patients, when available, with a particular focus on the presence or absence of late gadolinium enhancement (LGE) as an indicator of myocardial fibrosis. CMR was not obtained systematically across the full cohort and was reviewed only when available through routine clinical care. When possible, LGE location and pattern were also abstracted from imaging review and clinical documentation.

ICD Variables and Complications

Data on ICDs were collected through detailed chart reviews and device interrogation reports. For each patient, information was obtained regarding device type (including ICD, dual-chamber ICD, and cardiac resynchronization therapy defibrillator), indication for implantation (primary versus secondary prevention for ICDs), and any history of device-related therapy, including the occurrence and appropriateness of delivered shocks in ICD recipients. Additional ICD variables included age at implantation, device programming characteristics when available, and delivered therapies including antitachycardia pacing and shocks.

In addition, all recorded complications related to device implantation or function were documented, including lead-related issues, infections, pocket complications, and other adverse events. Follow-up data were reviewed to determine long-term outcomes and any subsequent interventions or revisions associated with device therapy.

Study Outcomes

The primary study outcome was documented sustained VT and VF in adults with MFS. Secondary analyses characterized ventricular arrhythmia phenotype, CMR findings in the subgroup with available advanced imaging, and ICD indications, therapies, and complications among ICD recipients.

Statistical Analysis

All statistical analyses were performed using Python 3.13. Data management and summary analyses were performed using pandas and NumPy; group comparisons were performed using SciPy; and logistic regression models were performed using statsmodels.

Categorical variables were summarized using frequencies and percentages. Continuous variables were assessed for normality as appropriate and reported as mean±SD or median with interquartile range, as appropriate. Baseline comparisons between patients with and without sustained VT/VF were performed using standard univariable comparisons according to variable type. Associations between clinical variables and the occurrence of VT or VF were evaluated using univariate logistic regression.

Covariates for the primary multivariable model were selected based on clinical relevance and representation of complementary domains, including demographics, ventricular ectopy, atrial arrhythmia burden, valvular disease, ventricular systolic function, and cardiac surgical history, rather than univariable significance alone. The primary multivariable logistic regression model included sex, age, NSVT, AF, moderate-severe MR, LVEF, and history of at least 1 cardiac surgery. Variables with greater missingness, including LV dimensions and LV mass index, were reported descriptively and in univariable analyses but were not included in the primary multivariable model.

A prespecified CMR sensitivity model was then performed in the subgroup with available CMR and included NSVT, moderate-severe MR, LVEF, and LGE. Analyses were performed using available case data for each variable, and model sample size therefore reflected complete data for included covariates. Univariable analyses were considered exploratory and hypothesis-generating, and no formal adjustment for multiple comparisons was performed. Because rhythm monitoring intensity and imaging availability were not standardized in this retrospective cohort, time-to-event modeling was not performed, and associations were instead assessed using logistic regression based on documented history of sustained VT/VF during available follow-up. Accordingly, the models were interpreted as association models based on documented history of sustained VT/VF rather than as prospective prediction models.

A 2-tailed $P < 0.05$ was considered statistically significant.

RESULTS

Study Population

The study included a total of 783 patients diagnosed with MFS (Table 1). The mean age was 51.3 ± 15.3 years, 443 (56.6%) were male, and 353 (45.1%) were genetically positive. Mean follow-up was 12.7 ± 7.6 years among 777 patients with available follow-up data. Sustained VT or VF occurred in 74 patients (9.5%).

Compared with patients without sustained VT/VF, those with sustained VT/VF were more often male (73.0% versus 54.9%; $P = 0.003$) and had higher all-cause mortality (36.5% versus 11.4%; $P < 0.001$), more stroke or transient ischemic attack (13.5% versus 6.8%; $P = 0.035$), diabetes (12.2% versus 5.2%; $P = 0.032$), chronic kidney disease (21.6% versus 10.6%; $P = 0.005$), AF (43.2% versus 22.8%; $P < 0.001$), atrial flutter (27.0% versus 11.7%; $P < 0.001$), supraventricular tachycardia (32.4% versus 15.8%; $P < 0.001$), NSVT (58.1% versus 7.3%; $P < 0.001$), abdominal aortic aneurysm (36.5% versus 21.4%; $P = 0.003$), type A aortic dissection (27.0% versus 13.3%; $P = 0.001$), and history of cardiac surgery (68.9% versus 55.1%; $P = 0.023$). They also had lower LVEF ($43.8 \pm 15.6\%$ versus $56.8 \pm 10.8\%$; $P < 0.001$), more elevated left atrial volume index (64.9% versus 31.1%; $P < 0.001$), more abnormal LV diastolic dimensions (46.4% versus 20.2%; $P < 0.001$), more abnormal LV systolic dimensions (57.1% versus 25.0%; $P < 0.001$), higher LV mass index (129.5 ± 56.3 versus 92.0 ± 28.7 g/m²; $P < 0.001$),

more moderate-severe MR (48.6% versus 19.9%; $P < 0.001$), MVP (55.4% versus 39.6%; $P = 0.009$), MAD (21.1% versus 11.9%; $P = 0.028$), and moderate-severe tricuspid regurgitation (40.5% versus 13.8%; $P < 0.001$; Table 1). Ascending aortic aneurysm was also more common in the sustained VT/VF group (90.4% versus 80.4%; $P = 0.036$), whereas age (54.4 ± 14.6 versus 51.0 ± 15.3 years; $P = 0.060$), genetic positivity (44.6% versus 45.1%; $P = 0.929$), hypertension (48.6% versus 43.9%; $P = 0.431$), obesity (10.8% versus 13.0%; $P = 0.595$), race, type B aortic dissection (16.2% versus 14.5%; $P = 0.696$), bundle branch block, moderate-severe aortic regurgitation (25.7% versus 17.8%; $P = 0.096$), and bicuspid aortic valve (6.8% versus 3.0%; $P = 0.090$) did not differ significantly (Table 1).

Ventricular Arrhythmia Phenotype and Detection

Among the 74 patients with sustained VT/VF, 60 (7.7% of the cohort) had sustained VT and 41 (5.2%) had VF; some patients experienced both arrhythmias (Table 1). In the broader ventricular arrhythmia cohort that included NSVT, sustained VT, or VF ($n = 126$), common NSVT detection modalities were Holter, loop, or event monitoring in 48 of 93 patients (51.6%) and ICD interrogation in 28 of 93 (30.1%). Sustained VT was commonly identified by ICD interrogation in 24 of 49 patients (49.0%), telemetry in 24 of 49 (49.0%), and ambulatory monitoring in 20 of 49 (40.8%). VF was most often identified through ICD interrogation in 19 of 29 patients (65.5%) or cardiac arrest presentation in 12 of 29 (41.4%) (Table S1).

When morphology was available, NSVT was most often monomorphic in 37 of 60 patients (61.7%), and sustained VT was also predominantly monomorphic in 19 of 31 patients (61.3%) (Table S1). Among adjudicated VF causes, unknown cause was most common in 9 of 38 events (23.7%), followed by cardiac surgical triggers in 4 of 38 (10.5%); cardiomyopathic, respiratory, ischemic, valvular, and QT-related mechanisms each accounted for smaller subsets (Table S1).

AF and Antiarrhythmic Therapy

AF was present in 194 patients (24.8%), including paroxysmal AF in 135 (17.2%) and persistent AF in 54 (6.9%) (Table 1). Overall use of β -blockers, calcium channel blockers, amiodarone, digoxin, flecainide, propafenone, dofetilide, dronedarone, and sotalol in the AF subgroup is summarized in Table S2.

CMR Imaging Subgroup

CMR data were available in 91 patients, including 15 patients with sustained VT/VF and 76 without sustained VT/VF. Age at CMR did not differ significantly between

Table 1. Baseline Characteristics Stratified by History of Sustained VT or VF

Characteristics	Overall (N=783)	Sustained VT or VF (N=74)	No sustained VT or VF (N=709)	P value
Age, y $\pm$ SD	51.3 $\pm$ 15.3	54.4 $\pm$ 14.6	51.0 $\pm$ 15.3	0.060
Gender				
Male, n (%)	443 (56.6%)	54 (73.0%)	389 (54.9%)	0.003
Female, n (%)	340 (43.4%)	20 (27.0%)	320 (45.1%)	0.003
Race				
White, n (%)	680 (86.8%)	65 (87.8%)	615 (86.7%)	0.791
Others, n (%)	103 (13.2%)	9 (12.2%)	94 (13.3%)	0.791
All-cause mortality, n (%)	108 (13.8%)	27 (36.5%)	81 (11.4%)	<0.001
Genetically positive, n (%)	353 (45.1%)	33 (44.6%)	320 (45.1%)	0.929
Follow-up duration, y $\pm$ SD	12.7 $\pm$ 7.6 (n=777)	12.5 $\pm$ 8.1	12.7 $\pm$ 7.5 (n=703)	0.853
Hypertension, n (%)	347 (44.3%)	36 (48.6%)	311 (43.9%)	0.431
Obesity, n (%)	100 (12.8%)	8 (10.8%)	92 (13.0%)	0.595
Stroke/transient ischemic attack, n (%)	58 (7.4%)	10 (13.5%)	48 (6.8%)	0.035
Diabetes, n (%)	46 (5.9%)	9 (12.2%)	37 (5.2%)	0.032
Chronic kidney disease, n (%)	91 (11.6%)	16 (21.6%)	75 (10.6%)	0.005
Arrhythmias				
Atrial fibrillation, n (%)	194 (24.8%)	32 (43.2%)	162 (22.8%)	<0.001
Paroxysmal, n (%)	135 (17.2%)	23 (31.1%)	112 (15.8%)	<0.001
Persistent, n (%)	54 (6.9%)	9 (12.2%)	45 (6.3%)	0.060
Atrial flutter, n (%)	103 (13.2%)	20 (27.0%)	83 (11.7%)	<0.001
Supraventricular tachycardia, n (%)	136 (17.4%)	24 (32.4%)	112 (15.8%)	<0.001
Nonsustained ventricular tachycardia, n (%)	95 (12.1%)	43 (58.1%)	52 (7.3%)	<0.001
Sustained VT or VF, n (%)	74 (9.5%)			
Sustained ventricular tachycardia, n (%)	60 (7.7%)			
Ventricular fibrillation, n (%)	41 (5.2%)			
Left bundle branch block, n (%)	54 (6.9%)	8 (10.8%)	46 (6.5%)	0.163
Right bundle branch block, n (%)	117 (14.9%)	14 (18.9%)	103 (14.5%)	0.313
ICD present, n (%)	56 (7.2%)	45 (60.8%)	11 (1.6%)	<0.001
Vascular				
Ascending aortic aneurysm, n (%)	635 (81.3%) (n=781)	66 (90.4%) (n=73)	569 (80.4%) (n=708)	0.036
Abdominal aortic aneurysm, n (%)	179 (22.9%)	27 (36.5%)	152 (21.4%)	0.003
Type A aortic dissection, n (%)	114 (14.6%)	20 (27.0%)	94 (13.3%)	0.001
Type B aortic dissection, n (%)	115 (14.7%)	12 (16.2%)	103 (14.5%)	0.696
Surgical				
History of at least 1 cardiac surgery, n (%)	442 (56.4%)	51 (68.9%)	391 (55.1%)	0.023
Aortic root surgery, n (%)	418 (53.4%)	55 (74.3%)	363 (51.2%)	<0.001
Aortic valve surgery, n (%)	310 (39.6%)	50 (67.6%)	260 (36.7%)	<0.001
Mitral valve surgery, n (%)	126 (16.1%)	30 (40.5%)	96 (13.5%)	<0.001
Tricuspid valve surgery, n (%)	20 (2.6%)	6 (8.1%)	14 (2.0%)	0.008
ASD/VSD closure, n (%)	9 (1.1%)	1 (1.4%)	8 (1.1%)	0.593
PFO closure, n (%)	37 (4.7%) (n=782)	5 (6.8%)	32 (4.5%) (n=708)	0.384
Structural				
LV ejection fraction, % $\pm$ SD	55.5 $\pm$ 11.9 (n=779)	43.8 $\pm$ 15.6	56.8 $\pm$ 10.8 (n=705)	<0.001
Elevated LAVI, n (%)	267 (34.3%) (n=778)	48 (64.9%)	219 (31.1%) (n=704)	<0.001
Abnormal diastolic LV dimension, n (%)	139 (23.2%) (n=599)	32 (46.4%) (n=69)	107 (20.2%) (n=530)	<0.001
Abnormal systolic LV dimension, n (%)	168 (28.4%) (n=592)	36 (57.1%) (n=63)	132 (25.0%) (n=529)	<0.001
LV mass index, g/m ² $\pm$ SD	96.3 $\pm$ 34.9 (n=550)	129.5 $\pm$ 56.3 (n=62)	92.0 $\pm$ 28.7 (n=488)	<0.001

(Continued)

Table 1. Continued

Characteristics	Overall (N=783)	Sustained VT or VF (N=74)	No sustained VT or VF (N=709)	P value
Moderate-severe mitral regurgitation, n (%)	177 (22.6%)	36 (48.6%)	141 (19.9%)	<0.001
Mitral valve prolapse, n (%)	322 (41.1%)	41 (55.4%)	281 (39.6%)	0.009
Mitral annular disjunction, n (%)	93 (12.8%) (n=725)	15 (21.1%) (n=71)	78 (11.9%) (n=654)	0.028
Moderate-severe aortic regurgitation, n (%)	145 (18.5%)	19 (25.7%)	126 (17.8%)	0.096
Bicuspid aortic valve, n (%)	26 (3.3%) (n=782)	5 (6.8%)	21 (3.0%) (n=708)	0.090
Moderate-severe tricuspid regurgitation, n (%)	128 (16.3%)	30 (40.5%)	98 (13.8%)	<0.001

Values are presented as mean±SD or n (%). Available-case denominators are shown where applicable. ASD indicates atrial septal defect; ICD, implantable cardioverter defibrillator; LAVI, left atrial volume index; LV, left ventricular; PFO, patent foramen ovale; VF, ventricular fibrillation; VSD, ventricular septal defect; and VT, ventricular tachycardia.

patients with and without sustained VT/VF (46.6 ± 11.4 versus 42.6 ± 14.4 years; $P=0.247$). LGE was present in 13 of 15 patients with sustained VT/VF (86.7%) compared with 14 of 76 patients without sustained VT/VF (18.4%; $P<0.001$; Table 2). Inferobasal and inferolateral involvement were common among patients with LGE and available location data, whereas septal LGE was numerically more frequent among patients with sustained VT/VF but did not reach statistical significance. LGE pattern distributions were otherwise similar between groups (Table 2). The Figure provides sample CMR images of LGE from our cohort.

Factors Associated With Sustained VT/VF

In univariate analysis, sustained VT/VF was associated with male sex (OR, 2.22 [95% CI, 1.30–3.79]; $P=0.003$), stroke or transient ischemic attack (OR, 2.15 [95% CI, 1.04–4.46]; $P=0.039$), diabetes (OR, 2.51 [95% CI, 1.16–5.44]; $P=0.019$), chronic kidney disease (OR, 2.33 [95% CI, 1.28–4.26]; $P=0.006$), AF (OR, 2.57 [95% CI, 1.57–4.21]; $P<0.001$), atrial flutter (OR, 2.79 [95% CI, 1.59–4.90]; $P<0.001$),

supraventricular tachycardia (OR, 2.56 [95% CI, 1.51–4.33]; $P<0.001$), NSVT (OR, 17.52 [95% CI, 10.20–30.10]; $P<0.001$), ascending aortic aneurysm (OR, 2.30 [95% CI, 1.03–5.13]; $P=0.041$), abdominal aortic aneurysm (OR, 2.11 [95% CI, 1.27–3.49]; $P=0.004$), type A aortic dissection (OR, 2.42 [95% CI, 1.39–4.23]; $P=0.002$), history of cardiac surgery (OR, 1.80 [95% CI, 1.08–3.02]; $P=0.025$), lower LVEF (OR, 0.94 per % [95% CI, 0.92–0.95]; $P<0.001$), elevated left atrial volume index (OR, 4.09 [95% CI, 2.47–6.76]; $P<0.001$), abnormal LV diastolic dimensions (OR, 3.42 [95% CI, 2.04–5.74]; $P<0.001$), abnormal LV systolic dimensions (OR, 4.01 [95% CI, 2.35–6.86]; $P<0.001$), higher LV mass index (OR, 1.02 per g/m² [95% CI, 1.02–1.03]; $P<0.001$), moderate-severe MR (OR, 3.82 [95% CI, 2.33–6.24]; $P<0.001$), MVP (OR, 1.89 [95% CI, 1.17–3.07]; $P=0.010$), MAD (OR, 1.98 [95% CI, 1.07–3.67]; $P=0.030$), moderate-severe tricuspid regurgitation (OR, 4.25 [95% CI, 2.55–7.08]; $P<0.001$), and LGE on CMR (OR, 28.79 [95% CI, 5.83–142.24]; $P<0.001$; Table S3).

In the primary multivariable model including 779 patients, NSVT (OR, 12.11 [95% CI, 6.43–22.79];

Table 2. Cardiac Magnetic Resonance Imaging Findings by History of Sustained VT VF

Characteristics	Overall (N=91)	Sustained VT or VF (N=15)	No sustained VT or VF (N=76)	P value
Age at CMR, y±SD	43.3±13.9 (n=90)	46.6±11.4	42.6±14.4 (n=75)	0.247
Late gadolinium enhancement, n (%)	27 (29.7%)	13 (86.7%)	14 (18.4%)	<0.001
Inferolateral, n (%)	17/26 (65.4%)	7/12 (58.3%)	10/14 (71.4%)	0.683
Inferobasal, n (%)	22/26 (84.6%)	9/12 (75.0%)	13/14 (92.9%)	0.306
Septal, n (%)	8/26 (30.8%)	6/12 (50.0%)	2/14 (14.3%)	0.090
Diffuse, n (%)	1/26 (3.8%)	1/12 (8.3%)	0/14 (0.0%)	0.462
LGE pattern (n=21)		(n=11)	(n=10)	
Midmyocardial, n (%)	8 (38.1%)	3 (27.3%)	5 (50.0%)	0.387
Subepicardial, n (%)	4 (19.0%)	3 (27.3%)	1 (10.0%)	0.586
Subendocardial, n (%)	3 (14.3%)	3 (27.3%)	0 (0.0%)	0.214
Transmural, n (%)	5 (23.8%)	3 (27.3%)	2 (20.0%)	1.000
Patchy or focal, n (%)	6 (28.6%)	3 (27.3%)	3 (30.0%)	1.000

LGE location and pattern categories were not mutually exclusive. Among 27 patients with LGE, location data were available in 26 patients, with 1 patient missing LGE location data; pattern data were available in 21 patients, with 6 patients missing LGE pattern data. Percentages for location and pattern rows were calculated using the available data for each respective variable. CMR indicates cardiac magnetic resonance; LGE, late gadolinium enhancement; VF, ventricular fibrillation; and VT, ventricular tachycardia.

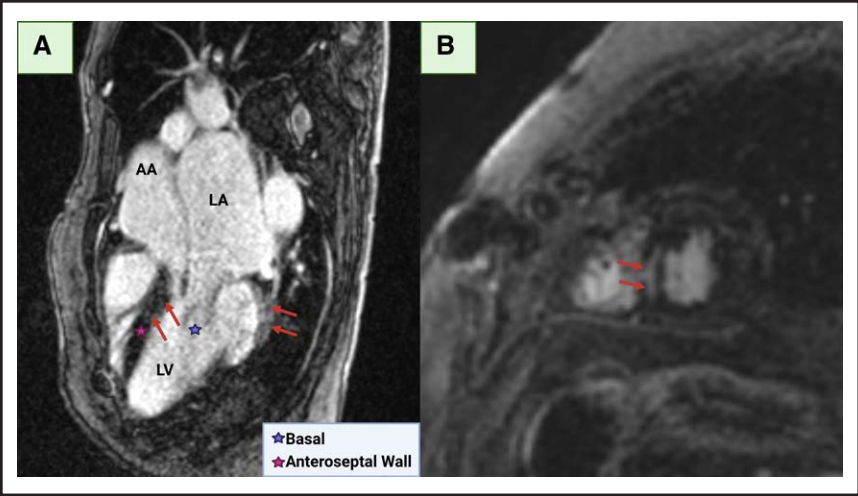


Figure. Representative cardiac magnetic resonance images demonstrating myocardial fibrosis by late gadolinium enhancement in patients with Marfan syndrome. **A**, Late gadolinium enhancement involving the basal to mid-septal segments, highlighted by red arrows. **B**, Mid-myocardial late gadolinium enhancement within the left ventricular (LV) septal myocardium, highlighted by red arrows. Myocardium, highlighted by red arrows. AA indicates ascending aorta; and LA, left atrium. Created in BioRender. Dreher, L. (2026) <https://BioRender.com/ff8w6fo>.

$P<0.001$), lower LVEF (OR, 0.95 per 1% increase [95% CI, 0.93–0.97]; $P<0.001$), history of at least 1 cardiac surgery (OR, 3.59 [95% CI, 1.65–7.84]; $P=0.001$), and moderate-severe MR (OR, 2.00 [95% CI, 1.07–3.75]; $P=0.030$) remained independently associated with sustained VT/VF, whereas male sex, age, and AF were not independently significant (Table 3).

In an exploratory CMR sensitivity model limited to 91 patients, NSVT (OR, 12.99 [95% CI, 2.12–79.52]; $P=0.006$), lower LVEF (OR, 0.94 [95% CI, 0.89–1.00]; $P=0.042$), and LGE (OR, 8.83 [95% CI, 1.37–56.97]; $P=0.022$) remained independently associated with sustained VT/VF, while moderate-severe MR did not (Table 4).

7 (14.9%), and cardiac resynchronization therapy defibrillator in 13 (27.7%).

Programming data were available in 46 patients. The mean VF detection zone was 206.0 ± 17.3 bpm, and the mean VT detection zone was 170.8 ± 20.3 bpm. Anti-tachycardia pacing followed by shock was the most common programmed therapy in both the VF zone (38/46, 82.6%) and VT zone (23/46, 50.0%). Among 41 patients with therapy data, 18 (43.9%) received anti-tachycardia pacing, and 17 (41.5%) received shocks. Any ICD complication occurred in 14 patients (25.0%), most commonly infection in 6 of 56 (10.7%) and lead malfunction in 5 of 56 (8.9%); 3 of 14 complications (21.4%) occurred within 1 month of implantation (Table S4).

ICD Recipients and Associated Outcomes

ICDs were present in 56 patients (7.2% of the cohort), including 45 of 74 patients (60.8%) with a history of sustained VT/VF (Table 1). Among ICD recipients, implantation occurred at a mean age of 46.4 ± 13.3 years, and indications were evenly divided between primary and secondary prevention (28 patients each; Table S4). Device type was available in 47 patients and included a single-chamber ICD in 27 (57.4%), dual-chamber ICD in

DISCUSSION

In this multicenter cohort of 783 adults with MFS, sustained VT or VF occurred in nearly 1 in 10 patients, emphasizing that ventricular arrhythmias represent a meaningful and likely underrecognized component of the cardiac phenotype in MFS. In the overall cohort, sustained VT or VF was independently associated with NSVT, lower ventricular ejection fraction, cardiac surgery history, and moderate-severe MR, suggesting that arrhythmic risk in

Table 3. Primary Multivariable Logistic Regression Model for Sustained Ventricular Tachycardia or Ventricular Fibrillation

Variables	OR (95% CI)	P value	N included
Sex (male)	1.73 (0.92–3.25)	0.091	779
Age (per 10 y)	0.99 (0.80–1.21)	0.894	779
Nonsustained ventricular tachycardia	12.11 (6.43–22.79)	<0.001	779
Atrial fibrillation	0.67 (0.33–1.37)	0.274	779
Moderate-severe mitral regurgitation	2.00 (1.07–3.75)	0.030	779
LV ejection fraction	0.95 (0.93–0.97)	<0.001	779
History of at least 1 cardiac surgery	3.59 (1.65–7.84)	0.001	779

Primary model covariates included sex, age per 10 y, nonsustained VT, atrial fibrillation, moderate-severe mitral regurgitation, LV ejection fraction, and history of at least 1 cardiac surgery. LV indicates left ventricular; and OR, odds ratio.

Table 4. CMR Sensitivity Multivariable Logistic Regression Model for Sustained Ventricular Tachycardia or Ventricular Fibrillation

Variables	OR (95% CI)	P value	N included
Nonsustained ventricular tachycardia	12.99 (2.12–79.52)	0.006	91
Moderate-severe mitral regurgitation	1.65 (0.29–9.28)	0.572	91
LV ejection fraction	0.94 (0.89–1.00)	0.042	91
Late gadolinium enhancement	8.83 (1.37–56.97)	0.022	91

Sensitivity model covariates included nonsustained ventricular tachycardia, moderate-severe mitral regurgitation, left ventricular ejection fraction, and late gadolinium enhancement. CMR indicates cardiac magnetic resonance; LV, left ventricular; and OR, odds ratio.

MFS is not defined by a single imaging or clinical marker but rather by the convergence of electrical instability, ventricular remodeling, prior operative burden, and valvular dysfunction. In the subgroup with CMR imaging, LGE remained independently associated with sustained VT or VF, supporting an additional role for myocardial scar in selected patients. Taken together, these findings extend prior observations of myocardial and arrhythmic involvement in MFS and support a broader framework for ventricular arrhythmia risk stratification that moves beyond the aorta alone.

Prior studies have suggested that ventricular arrhythmias in MFS occur more often than traditionally appreciated, but the reported burden has varied widely across cohorts, reflecting differences in study size, ascertainment, imaging availability, and definitions of arrhythmic end points.^{3–6} Earlier series identified NSVT and sustained ventricular arrhythmias in a subset of patients with MFS and raised the possibility that myocardial involvement extends beyond the aortic root and valve apparatus alone.^{3–6} Our findings build on that literature by examining ventricular arrhythmias within a larger multicenter cohort while integrating clinical history, echocardiographic markers of remodeling and valvular disease, surgical history, and CMR data when available. In doing so, this study helps place ventricular arrhythmias within a broader Marfan cardiac phenotype and suggests that risk is shaped by the combined burden of electrical, structural, and postoperative factors rather than by any single abnormality in isolation.^{3–7}

NSVT emerged as one of the strongest markers of sustained ventricular arrhythmia in this cohort, suggesting that in MFS it may reflect a meaningful arrhythmogenic substrate rather than an incidental finding. This interpretation is supported by prior reports linking ventricular ectopy and NSVT with more advanced myocardial involvement and adverse cardiac outcomes in MFS.^{3,5,6} At the same time, NSVT ascertainment in a retrospective cohort is inherently influenced by monitoring intensity. Patients with more advanced disease, cardiac surgery history, ICDs, or greater clinical concern are more likely to undergo ambulatory monitoring, telemetry, or device interrogation, increasing the likelihood that NSVT will be detected. In that sense, NSVT may function both as

a marker of electrical instability and as a surrogate for a subgroup already recognized clinically as higher risk. Even with that caveat, its strong association with sustained VT/VF in the present study suggests that NSVT should not be dismissed in MFS and instead may warrant closer rhythm surveillance and more careful structural evaluation.

Lower LVEF was also independently associated with sustained VT/VF, pointing to a broader pattern of ventricular remodeling in higher-risk patients. This relationship was not limited to systolic function alone. In univariate analysis, sustained VT/VF was also associated with abnormal LV diastolic and systolic dimensions as well as higher LV mass index, suggesting that arrhythmic risk in MFS may accompany progressive structural remodeling rather than depend on a single threshold of dysfunction.^{10,11} These findings are consistent with prior work showing that myocardial involvement in MFS can extend beyond valvular or aortic pathology and may include intrinsic ventricular disease with potential arrhythmic consequences.^{3,4} Taken together, the association of reduced ejection fraction with other markers of adverse chamber remodeling supports the idea that ventricular arrhythmias in MFS often arise in the setting of a more globally abnormal myocardial phenotype.

Moderate-severe MR remained independently associated with sustained VT/VF in the primary model, underscoring the importance of mitral valve disease within the broader arrhythmic phenotype of MFS. This signal was accompanied by higher rates of MVP and MAD on baseline and univariate analyses. Prior studies in MFS have similarly linked mitral valve involvement, particularly MAD, with adverse cardiovascular outcomes and a more pronounced myocardial phenotype.⁸ More broadly, abnormal valvular mechanics may contribute to ventricular irritability through chronic volume loading, altered annular dynamics, papillary muscle stress, and localized remodeling, all of which may lower the threshold for malignant ventricular arrhythmias.^{8,9,12}

History of cardiac surgery also remained independently associated with sustained VT/VF, identifying a subgroup with a greater overall burden of cardiovascular disease. This finding should be interpreted

cautiously, as surgery in a retrospective cohort likely reflects several overlapping processes rather than a single mechanism. In our analysis, surgical history was defined according to procedures occurring before sustained VT/VF among those with events, or as ever having undergone cardiac surgery among those without a history of sustained VT/VF, allowing operative history to be interpreted in a clinically relevant temporal context. Notably, ventricular arrhythmic events were not confined to advanced age alone; among patients with VF, the mean time to first event was 43.1 ± 14.3 years, suggesting that malignant ventricular arrhythmias may occur relatively early in a subset of patients with MFS. Patients requiring aortic or valvular intervention at younger ages may therefore represent a particularly vulnerable subgroup, although more intensive surveillance in these patients may also increase arrhythmia detection.¹³ Even with these caveats, the persistence of cardiac surgery history in the primary model suggests that operative burden identifies a clinically higher-risk subgroup in MFS and should be considered alongside rhythm, ventricular, and valvular features when evaluating ventricular arrhythmic risk.

In the subgroup that underwent CMR imaging, LGE remained independently associated with sustained VT/VF, supporting a role for myocardial scar or focal fibrosis in the arrhythmic phenotype of MFS.¹⁴ This finding adds important context to the broader full-cohort analysis.¹⁵ In the overall population, risk was not explained by CMR findings alone, but among patients selected for advanced imaging, LGE appeared to identify a subgroup with additional myocardial vulnerability. That distinction is important, particularly in a retrospective cohort where CMR was not performed uniformly and was more likely to be obtained in patients with greater clinical concern or a more complex phenotype. Prior studies have suggested that myocardial involvement in MFS may extend beyond the aorta and valves, and our findings further support the view that tissue-level myocardial abnormalities may contribute to malignant ventricular arrhythmias in selected patients.^{3,4} Although CMR may not function as a universal screening tool for all patients, it should remain a cornerstone of risk stratification when cardiac phenotypic involvement is suspected, especially when scar-mediated arrhythmic substrate is clinically relevant.

Taken together, our findings suggest that sustained VT/VF in MFS tends to occur within a recognizable high-risk phenotype marked by electrical instability, ventricular remodeling, valvular dysfunction, prior operative history, and myocardial scar. Rather than treating ventricular arrhythmias in MFS as rare exceptions, this study suggests that a subset of patients may be identifiable before malignant arrhythmias occur through a broader assessment of rhythm history, chamber and valvular abnormalities, surgical burden, and advanced imaging when appropriate. Just as importantly, the present results raise

several questions for prospective study, including when these abnormalities first emerge, whether they develop in parallel or sequentially, and which features most closely precede sustained VT/VF. Clarifying how NSVT, ventricular or valvular remodeling, surgical intervention, and myocardial scar relate to later ventricular arrhythmias may help create a more practical framework for surveillance and risk stratification. Although these findings do not establish a new risk stratification algorithm, they do suggest that ventricular arrhythmic risk in MFS may be more recognizable, and potentially more actionable, than has traditionally been appreciated.

Limitations

Several limitations merit consideration. First, this was a retrospective study from tertiary referral centers and is therefore subject to referral bias, ascertainment bias, and residual confounding. Detection of ventricular arrhythmias, particularly NSVT, was influenced by monitoring intensity, and patients with more advanced disease, ICDs, or greater clinical concern were more likely to undergo ambulatory monitoring, telemetry, or device interrogation. Therefore, the observed prevalence of NSVT and sustained VT/VF may be influenced by differential surveillance intensity and should not be interpreted as a population-level incidence estimate. Second, selected echocardiographic and surgical variables were not uniformly measured before arrhythmia onset in all patients. As a result, the present analysis cannot determine whether these features preceded VT/VF or instead reflect a more advanced structural-electrical phenotype among patients with documented sustained ventricular arrhythmias. Third, missingness was present for selected echocardiographic variables, particularly LV dimensions and LV mass index; these variables were therefore interpreted descriptively rather than as primary drivers of the multivariable conclusions. Fourth, given the number of univariable comparisons, statistically significant univariable findings should be interpreted as exploratory and hypothesis-generating. Fifth, CMR was available only in a subset of patients, introducing the possibility of selection bias in analyses involving LGE, and the exploratory CMR sensitivity model was limited by the small number of patients with CMR data and sustained VT/VF events. Finally, logistic regression does not account for the timing of VT/VF events during follow-up and treats events occurring at different time points similarly. Accordingly, the findings should be interpreted as associations with documented sustained VT/VF history rather than time-dependent risk estimates. Even with these limitations, the multicenter design and detailed integration of arrhythmic, structural, surgical, CMR, and ICD variables provide a comprehensive view of ventricular arrhythmic burden in adults with MFS.

Conclusions

Sustained VT/VF affects a clinically meaningful subset of adults with MFS and appears to arise within a broader structural-electrical phenotype. In the full cohort, NSVT, lower LVEF, history of cardiac surgery, and moderate-severe MR were independently associated with sustained VT/VF, while LGE provided additional discriminatory value among patients who underwent CMR. These findings reinforce that ventricular arrhythmic risk in MFS extends beyond aortic disease alone and support a more comprehensive framework for identifying higher-risk patients. Prospective studies will be needed to refine surveillance strategies, clarify the role of advanced imaging, and determine how best to translate these signals into individualized arrhythmic risk stratification.

ARTICLE INFORMATION

Received February 4, 2026; accepted September 2, 2026.

Created in BioRender. Dreher, L. (2026) <https://BioRender.com/k8vfmvcv>.

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Sources of Funding

None.

Disclosures

None.

Supplemental Material

Tables S1–S4

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